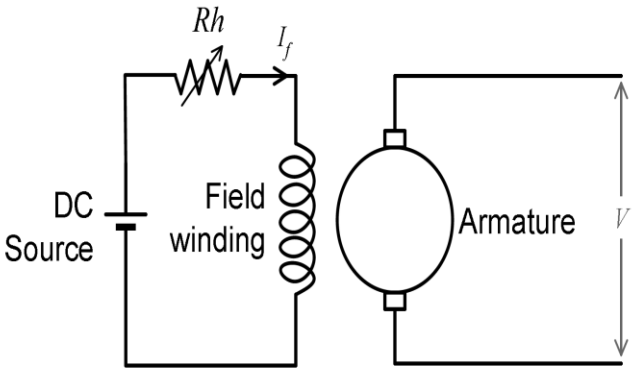
Module 1
DC Machines

- **DC shunt generator - no-load (open circuit) characteristics - critical field resistance, critical speed, voltage buildup –**
- **load characteristics – armature reaction - cross magnetising & demagnetising effect (concepts only)**
- **Power flow diagram – losses and efficiency – maximum efficiency –numerical problems**

Open circuit characteristics of shunt generator

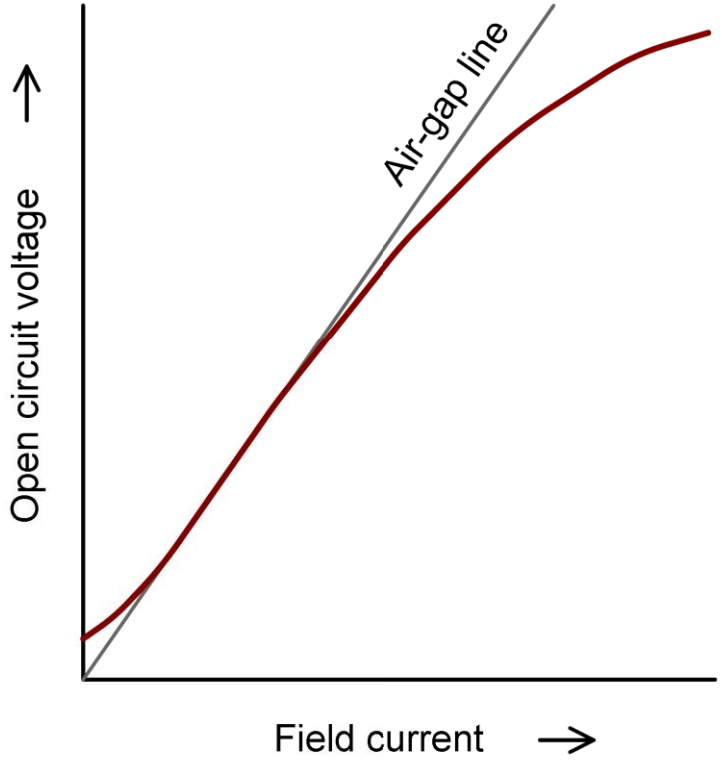
How is OCC plotted?

- The field winding is excited from a DC source and the field current is varied in steps by adjusting the rheostat connected in series
- The generated voltage across the armature terminals is recorded for each value of field current and the curve is plotted
- OCC is the plot of terminal voltage of the DC generator against field current and without any load connected to it



Open circuit characteristics of shunt generator

- The characteristics make a curve at higher values of field current due to the magnetic saturation of the core material.
- There is a small voltage induced even when field current is zero. This is due to the residual magnetism in the core
- The tangent drawn at the lower part of the OCC curve is called air-gap line
- Airgap line represents the mmf required to overcome the reluctance of the airgap



Terms associated with OCC

- **Residual voltage**

  Small voltage induced in the armature when field current is zero

- **Critical resistance**

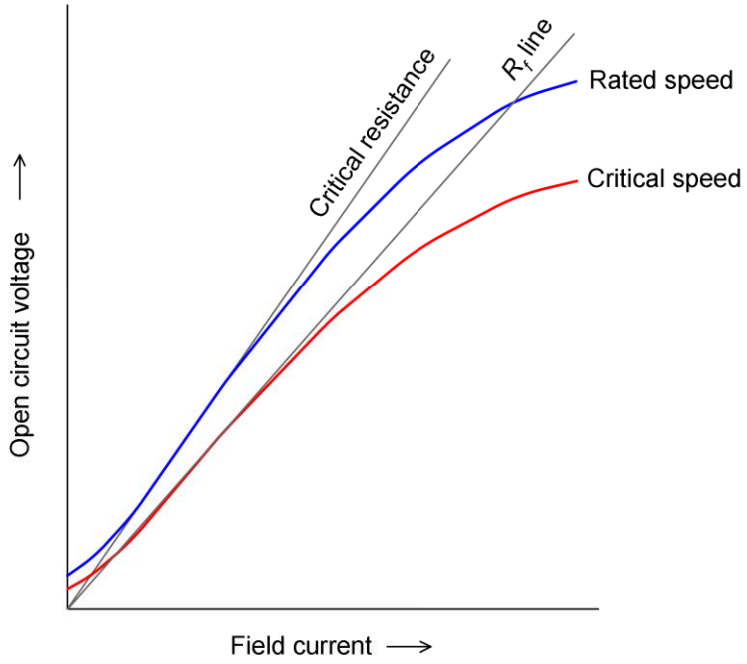
  The resistance of the field circuit above which the shunt generator will fail to excite at rated speed operation

- **R_f line**

  Line representing the resistance of the field winding without any external resistance

- **Critical speed**

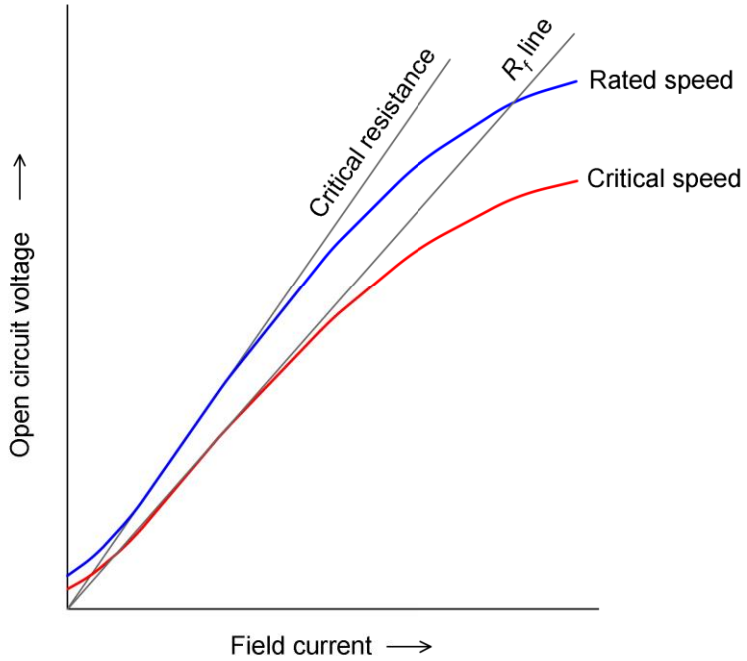
  Speed below which a dc shunt machine will fail excite even at minimum resistance in field circuit



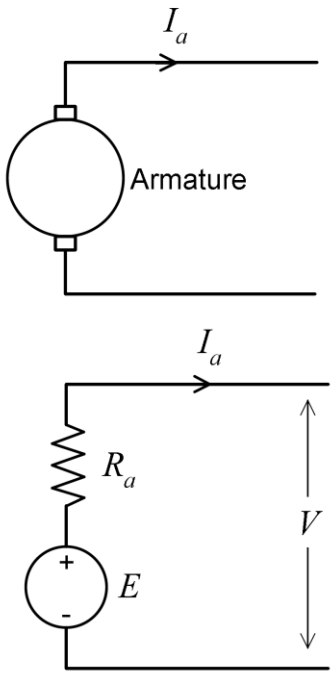
Conditions for voltage build up in a DC shunt generator

Essential conditions for voltage buildup in dc shunt generators

- Residual magnetism should be present in the pole core
- The connections of the armature and field windings should be in such a way that the voltage induced due to residual magnetism will further increase the flux in the pole
- The field circuit resistance should not be above the critical resistance
- The speed of the machine should be above the critical speed



Load Characteristics



The terminal voltage of the generator is given by

$$V = E - I_a R_a$$

where V = terminal voltage
E = induced voltage
I_a = armature current
R_a = armature resistance

For series and compound generators

$$V = E - I_a \left(R_a + R_s \right)$$

R_s = resistance of the series field winding

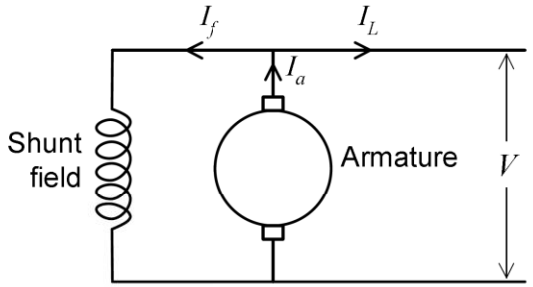
Example 2.1

A DC shunt generator has a terminal voltage of 220 V, field current of 0.5 A, and load current of 12 A. Find the induced voltage if the armature resistance is 0.2 Ω

Armature current, $I_a = I_L + I_f = 12 + 0.5 = 12.5$ A

We have $V = E - I_a R_a$

Then $E = V + I_a R_a = 220 + 12.5 \times 0.2 = \mathbf{222.5\ V}$



Example 2.2

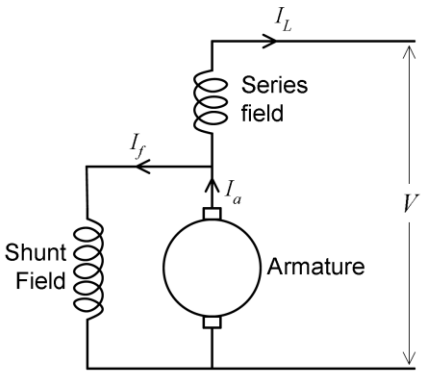
A compound wound dc generator with short shunt connection has an armature resistance of 0.15 Ω and a series field winding resistance of 0.1 Ω. The terminal voltage is 220 V when the machine delivers a current of 21A. If the shunt field current is 1 A, find the induced voltage.

Armature current, $I_a = 21 + 1 = 22$ A

Voltage drop in armature = $22 \times 0.15 = 3.3$ V

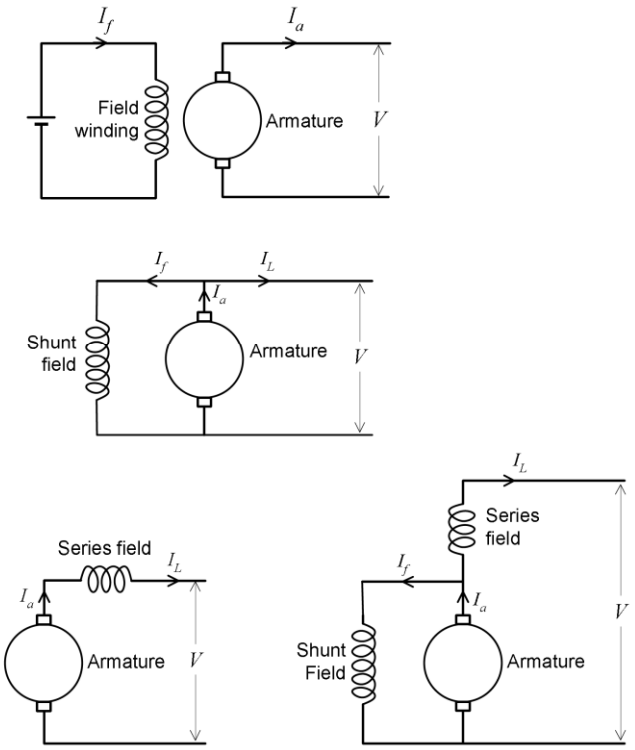
Voltage drop across series field winding = $21 \times 0.1 = 2.1$ V

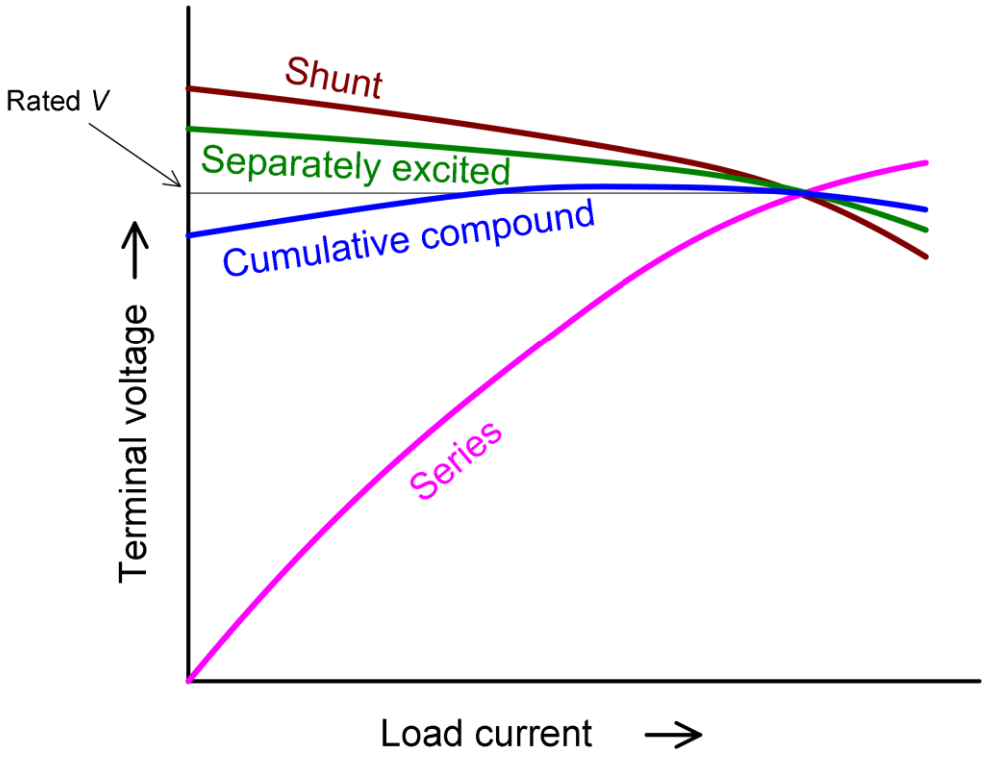
Induced voltage = $220 + 3.3 + 2.1 =$ **225.4 V**



$$E = V + I_a R_a + I_s R_s$$

Generator characteristics





Example 2.3

A dc shunt generator has the following characteristics when operating at its rated speed of 1500 rpm

I_f	0	0.1	0.2	0.3	0.4	0.5	0.6	0.8	1.0
E	6	50	101	152	187	212	229	245	251

The field winding resistance is 325 ohms. Find the following

a) The voltage induced with no additional resistance in field circuit
b) Resistance to be added to generate open circuit voltage of 220V
c) Critical resistance
d) Critical speed

Steps:

- Plot the OCC corresponding to 1500 rpm based on the given data
- Plot the R_f line – find (a) from the graph
- Find I_f corresponding to 220 V from the graph – find field total field circuit resistance – find resistance to be added
- Plot tangent to OCC to get critical resistance. Also calculate critical speed from graph

a Draw the R_f line and from the point it cuts the OCC obtain the voltage as 240 V

b Field current corresponding to $220\text{V} = 0.55\ \text{A}$

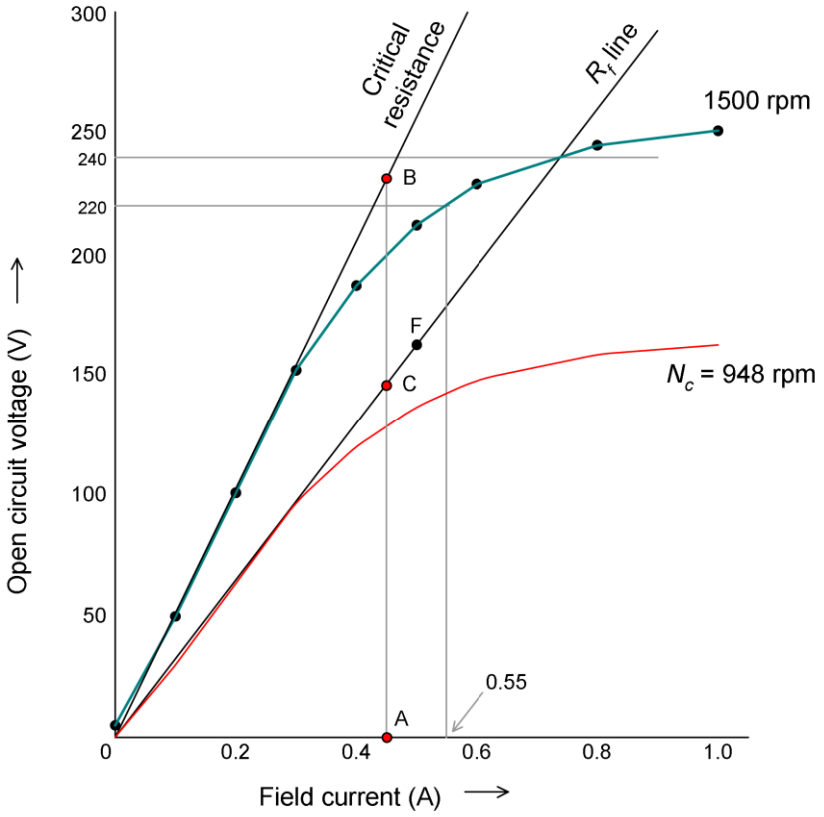
$$\text{Resistance of field circuit} = \frac{220}{0.55} = 400\ \Omega$$

$$\text{Additional resistance} = 400 - 325 = 75\ \Omega$$

c
$$\text{Critical resistance} = \frac{AB}{OA} = \frac{231}{0.45} = 513\ \Omega$$

d
$$\text{Critical speed} = \text{Rated speed} \times \frac{AC}{AB}$$

$$= 1500 \times \frac{146}{231} = 948\ \text{rpm}$$



Example 2.4

A long shunt DC compound generator delivers 25 kW power at 500 V. The resistance of armature, series and shunt winding are 0.05, 0.03 and 200 ohms respectively. Calculate generated emf and armature current. Assume 1 V per brush for contact drop.

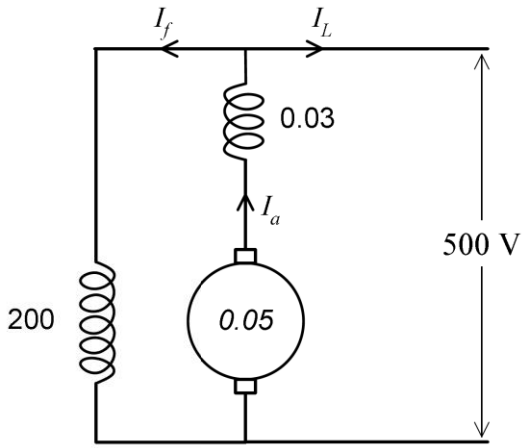
Load voltage, $V_L = 500$ V Power, $P = 25$ kW

Load current, $I_L = \frac{P}{V_L} = \frac{25 \times 10^3}{500} = 50$ A

Field current, $I_f = \frac{V_L}{R_f} = \frac{500}{200} = 2.5$ A

Armature current, $I_a = I_L + I_f = 50 + 2.5 = 52.5$ A

Generated voltage, $E = 500 + 52.5 \times (0.05 + 0.03) + 2 \times 1 = 506.2$ V



Example 2.5

Two DC shunt generators with induced emfs of 230V and 225V, armature resistance of 0.5 Ω and 0.4 Ω and field resistances of 100 Ω and 110 Ω respectively are in parallel with a terminal voltage of 220V. Calculate the power delivered by each generator.

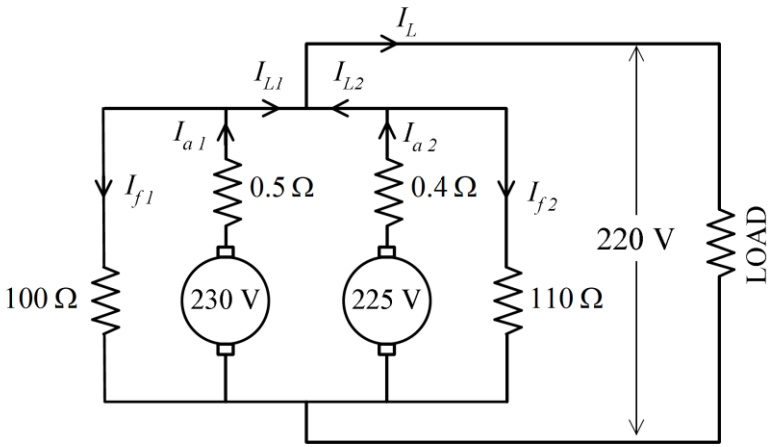
Field current of the first generator,

$$I_{f1} = \frac{220}{100} = 2.2 \text{ A}$$

Field current of the second generator,

$$I_{f2} = \frac{220}{110} = 2 \text{ A}$$

Equivalent circuit:



For the first generator, $220 + (I_{L1} + 2.2) \times 0.5 = 230$

$$(I_{L1} + 2.2) \times 0.5 = 10$$

$$I_{L1} = 17.8 \text{ A}$$

$\because \quad V + I_a \times R_a = E$

For the second generator, $220 + (I_{L2} + 2) \times 0.4 = 225$

$$(I_{L1} + 2) \times 0.4 = 5$$

$$I_{L2} = 10.5 \text{ A}$$

Power delivered by the first generator, $P_1 = V \times I_{L1}$

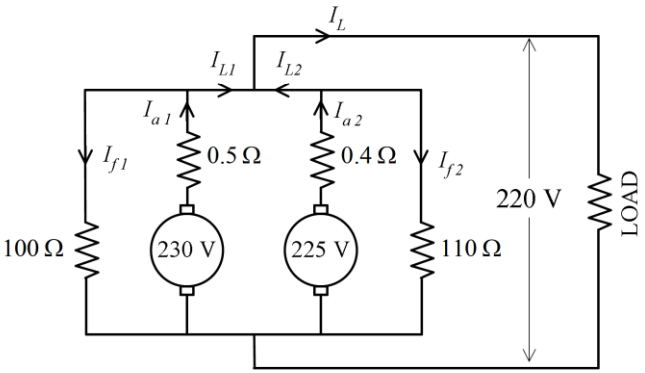
$$= 220 \times 17.8$$

$$= 3.916 \text{ kW}$$

Power delivered by the second generator, $P_2 = V \times I_{L2}$

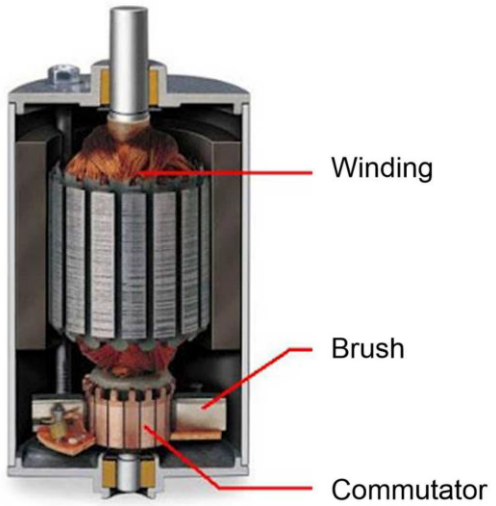
$$= 220 \times 10.5$$

$$= 2.31 \text{ kW}$$



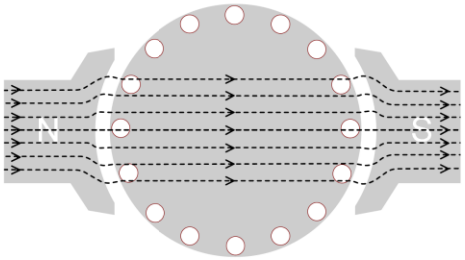
Commutation

- The reversal of current in the armature coil by brush and commutator is called commutation
- The reversal of current takes place while the coil passes through the magnetic neutral axis.
- The coil is short circuited through the brush during commutation
- Time of short circuit is called period of commutation
- Poor commutation may cause sparking at the brushes



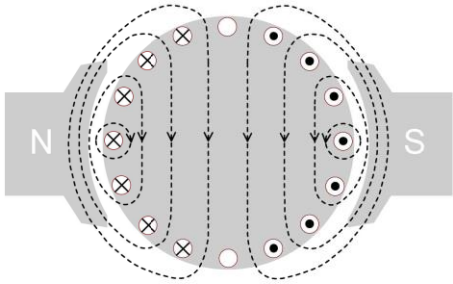
Armature reaction

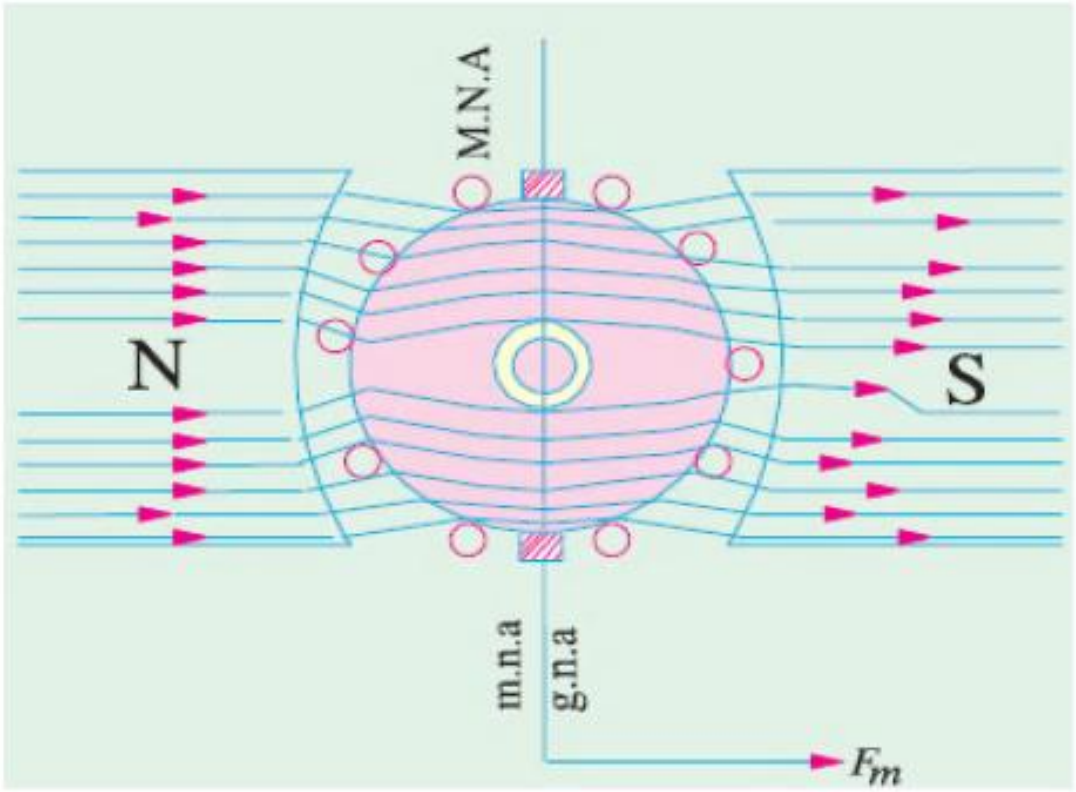
The effect of armature mmf on the main field flux distribution in the airgap is called armature reaction

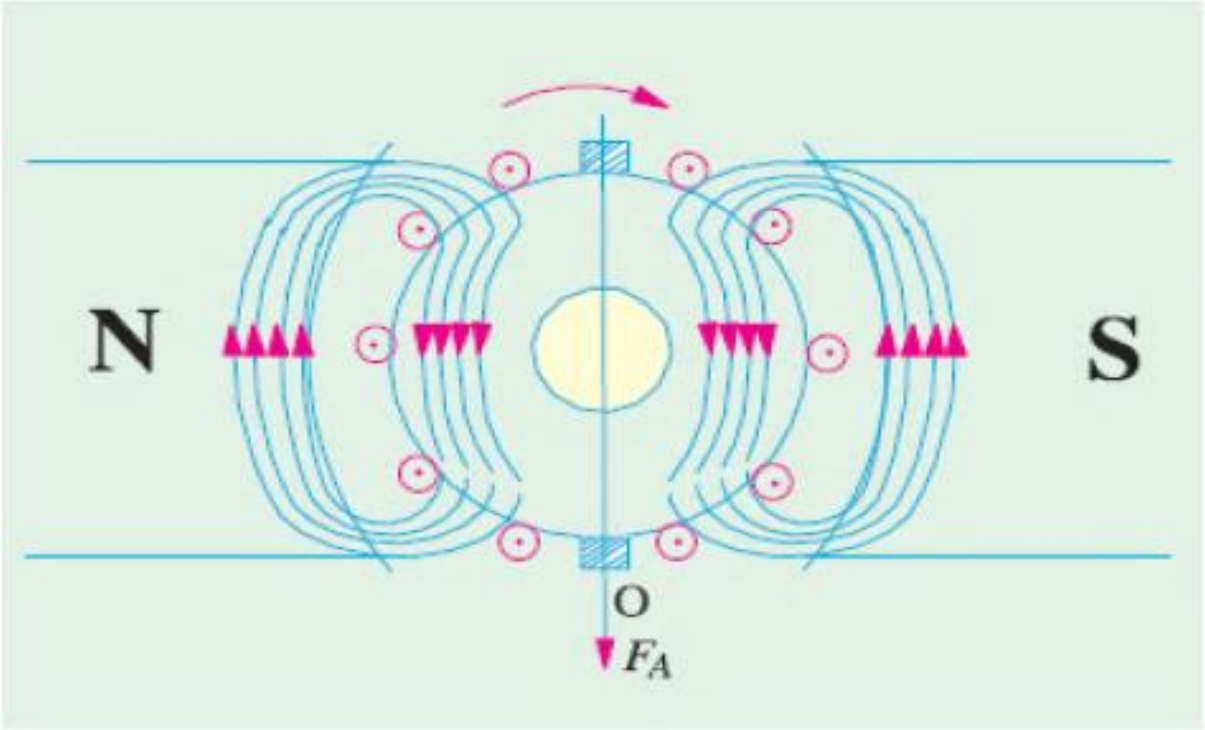


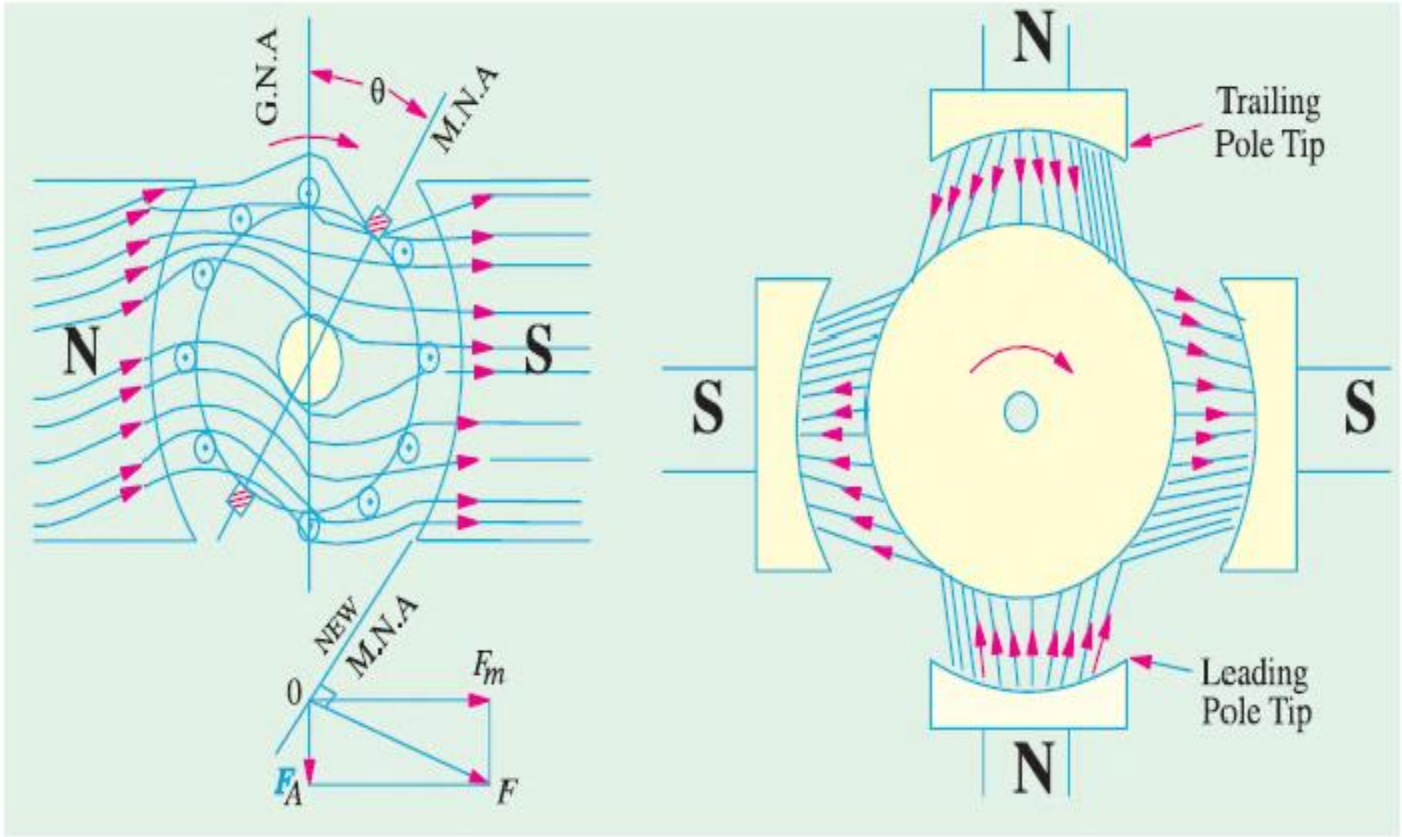
Effect of armature reaction

- Net reduction in per pole flux
- Distortion in the field flux wave along the airgap periphery
- Shifting of magnetic neutral axis



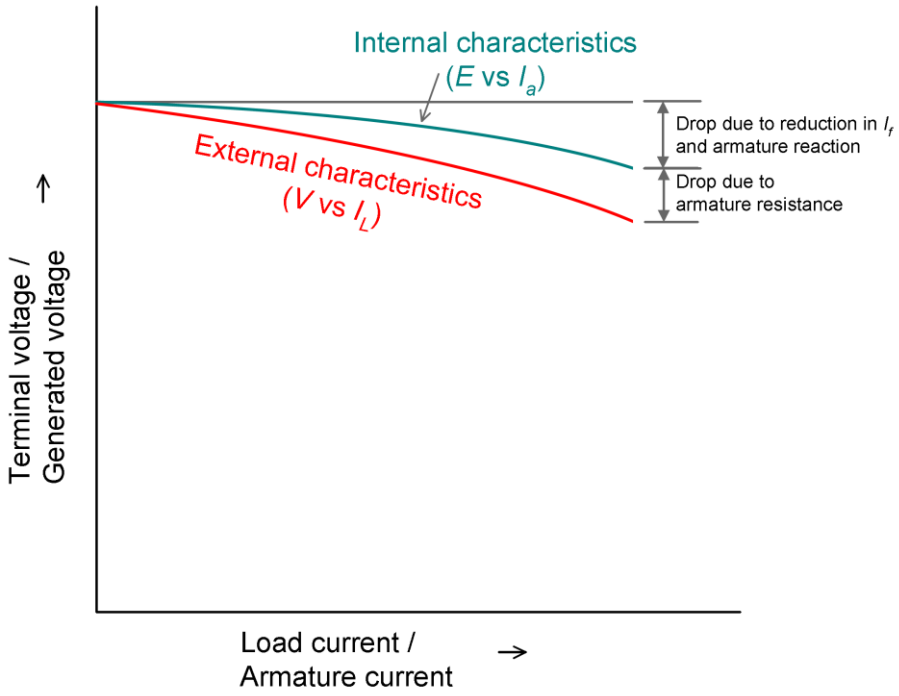

M.N.A
N
S
m.n.a
g.n.a
F_m


N
S
O
F_A

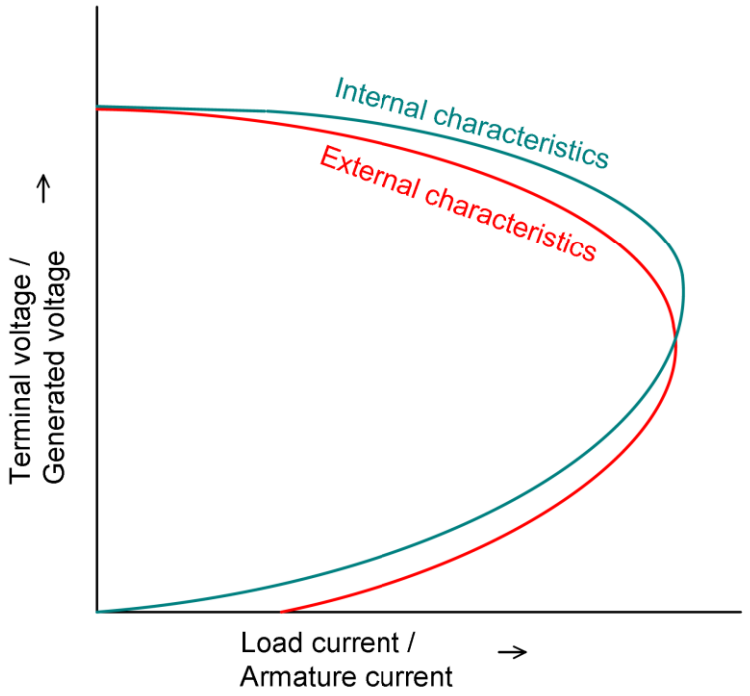

G.N.A
θ
M.N.A
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S
NEW
M.N.A
0
F_m
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F
N
Trailing
Pole Tip
S
S
Leading
Pole Tip
N

Load characteristics of shunt generator

Characteristics upto rated load

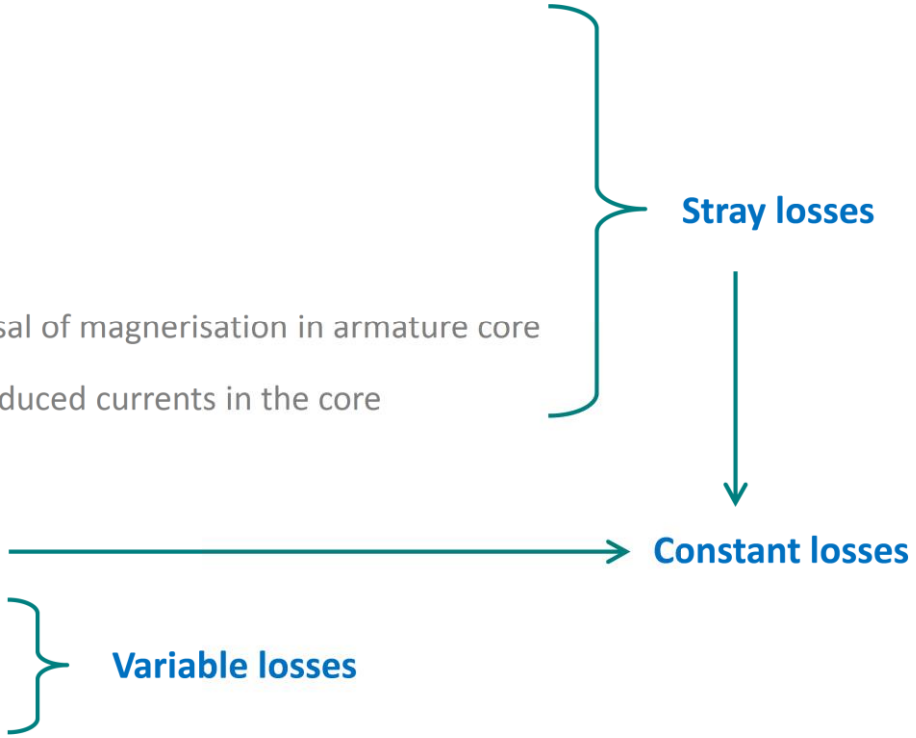


Complete characteristics



Losses in DC generator

- Mechanical loss
  - Friction loss
  - Windage loss
- Iron loss
  - Hysterisis loss – due to reversal of magnerisation in armature core
  - Eddy current loss – due to induced currents in the core
- Copper loss
  - Shunt field copper loss
  - Series field copper loss
  - Armature copper loss



The total losses in a dc machine are

Cupper losses are mainly due to the current passing through the winding.

Cu losses
- 1.Armature cu losses (30 to 40% of full load losses)
- 2.Shunt field cu losses(20 to30% of full load losses)
- 3.Series field cu losses

Armature cu losses=$I_a^2 R_a$

R_a=Armature resistance

I_a= Armature current

--Losses due to brush contact resistance is usually include in armature cu losses

Shunt field cu losses=$I_{sh}^2 R_{sh}$

R_{sh}=Shunt field resistance

I_{sh}=Shunt field current

Series field cu losses=$I_{se}^2 R_{se}$

R_{se}=Series field resistance

I_{se}=Series field current

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Iron losses (Magnetic losses) (20 to 30% of full load losses)

1)Hysteresis losses

2)Eddy current losses

Hysteresis losses (W_h):-

The losses is due to the reversal of magnetisation of the armature core

Every portion of the rating core passes under N and S poles alternately. There by attaining S and N polarity respectively. The core undergoes one complete cycle of magnetic reversal after passing under one pair of poles.

P=No. of poles

N= Armature speed in rpm

frequency of magnetic reversals

$$f=\frac{NP}{120}$$

The losses depends upon the volume and B_{max} and frequency of reversals.

Hysteresis losses is given by steinmetz formula

$$W_h=\eta B^{1.6}_{max} f V \text{ wats}$$

V=Volume of the core in m^3

η= Steinmetz hysteresis coefficient

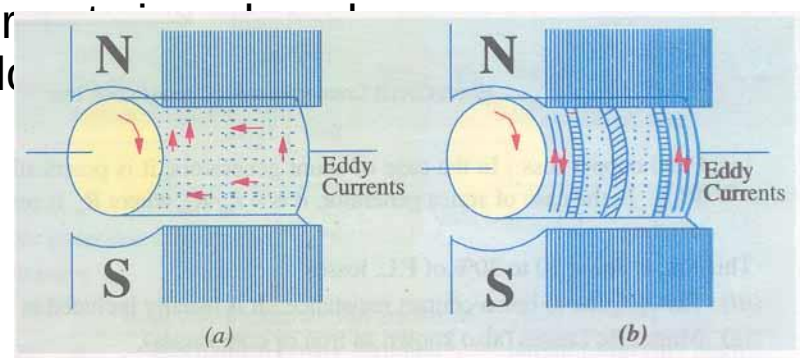
Eddy current losses:-(W_e)

when the armature core rotates, it cuts the magenetic flux hence an e.m.f induced in in the body of the core according to faradays law of electro magnetic induction. This e. m.f through small sets up large current in the body of the core due to its mall resistance. This current is known as "Eddy Current"

-These core laminations are insulated from each other by a thin coating of varnish. Due to the core body being one continuous solid iron piece (fig a)

The magnitude of eddy current is large. As armature cross sectional area is large it's resistance is small. hence eddy current losses is large.

In (fig b) The same core has been split up in to thin cross section has very high resistance, hence magnitude of eddy cur considerably there by reducing eddy current l



$W_e = k\, B^2_{max} f^2 t^2 v^2$ watts

B_{max}=maximum flux densities

f=Freequency of the magenetic reversals

v=volume of the armaturecore

t=Thick ness of lamination

$w_e \propto t^2$ hence t should be kept as small as posible.

Eddy current losses is reduced by laminated core but hysteresis losses can not be reduced by this way.

Mechanical losses (10 to 20% of full load losses)

1.Friction losses

2.Windage losses

Friction losses:-

Frictional losses due to bearings

Windage losses:- Windage losses due to air gap between armature and pole shoe

Stray losses(Rotational losses):-

magnetic losses and mechanical losses are collectively known as stray losses

Losses are classified in to two types:-

i) Constant losses (standing losses)(W_c)

--Field cu losses is constant

--for shunt and compound generator are constant losses

so, stray losses+ shunt cu losses are combined called "constant losses"

ii) Variable losses:-The losses which varies with the load called "variable losses"

-- Armature cu losses is know as "variable losses"

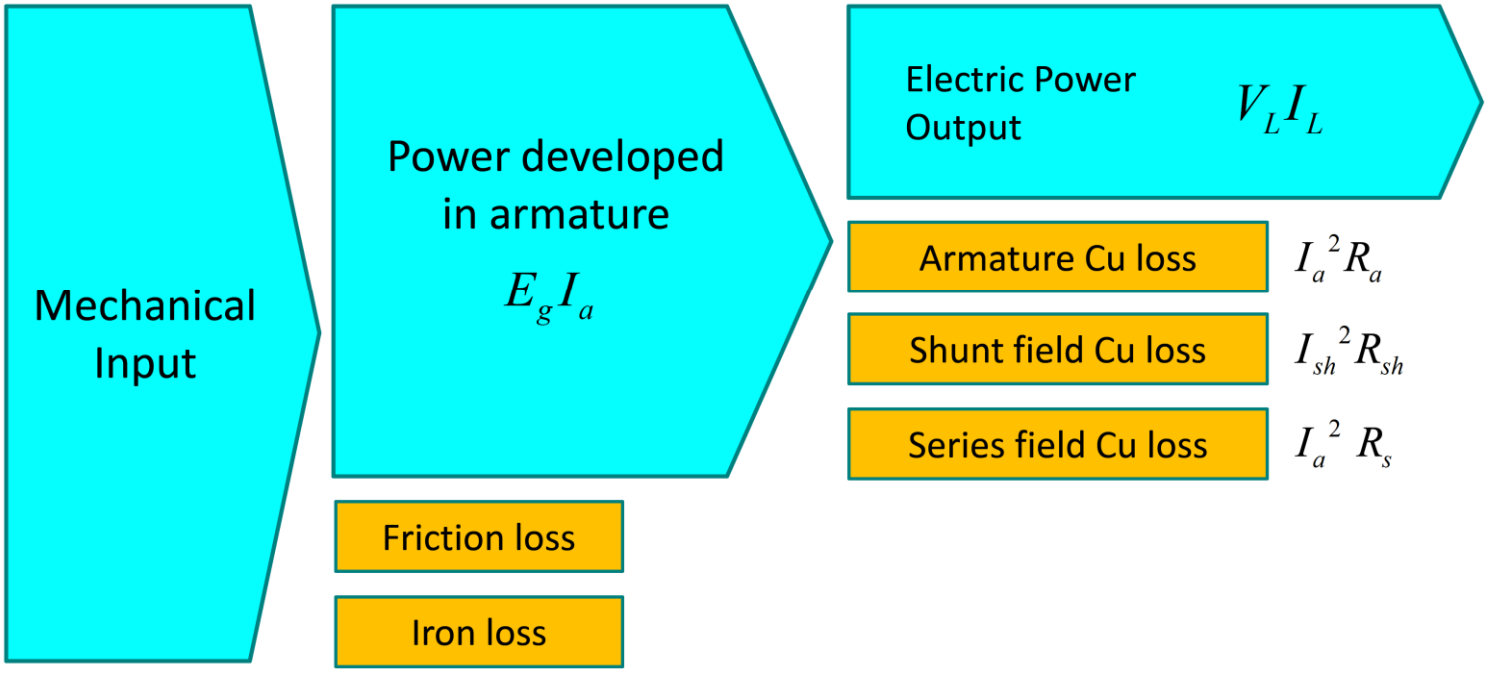
-- In series generator shunt field cu losses also variable losses ($I_L=I_{se}=I_a$)

So, Total losses=Armature copper losses + W_C

$=I_a^2R_a+W_c=(I+I_{sh})^2R_a+W_c$

Total losses=Variable losses+ Constant losses

Power flow in DC generator

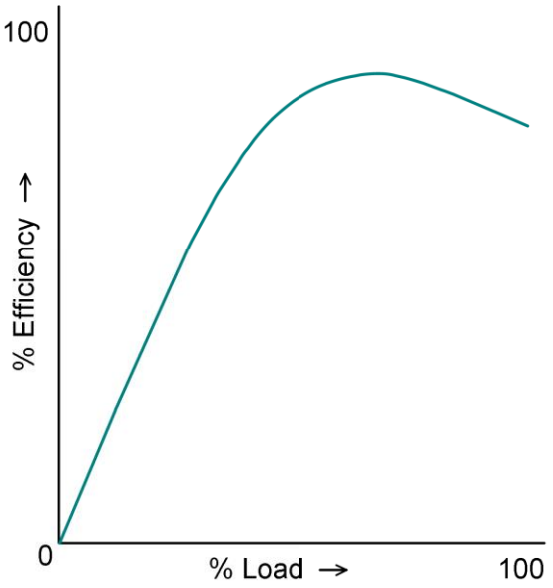


Efficiency of DC generator

Overall efficiency of the generator,

$$\eta = \frac{\text{Electrical output}}{\text{Mechanical input}}$$

$$= \frac{\text{Electrical output}}{\text{Electrical output} + \text{losses}}$$



Condition for maximum efficiency

Generator output $= VI$

Generator input $= VI + \text{losses}$

$= VI + I^2 R_a + W_c$

Assuming $I = I_a$ since I_f is negligible

$W_c =$ Constant losses

$$\text{Efficiency, } \eta = \frac{Output}{\text{Input}} = \frac{VI}{VI + I^2 R_a + W_c} = \frac{1}{1 + \left(\frac{IR_a}{V} + \frac{W_C}{VI} \right)}$$

For maximum efficiency, the denominator should be minimum. i.e. $\frac{d}{dI}\left(\frac{IR_a}{V} + \frac{W_C}{VI} \right) = 0$

$$\frac{d}{dI}\left(\frac{IR_a}{V} + \frac{W_C}{VI} \right) = \frac{R_a}{V} - \frac{W_C}{VI^2} = 0 \Rightarrow \frac{R_a}{V} = \frac{W_C}{VI^2} \Rightarrow I^2 R_a = W_C$$

Condition for maximum efficiency Variable loss = Fixed loss

Example

A sunt generator supplies a full load current of 200 A at a terminal voltage of 250 V. The armature resistance and field resistance are 0.08 Ω and 50 Ω respectively. The iron and friction losses are 900 W. Find

a) Cu losses at full load
b) Efficiency at full load
c) Maximum Efficiency
d) Load current at maximum efficiency

Shunt field current, $I_{sh} = \frac{250}{50} = 5\text{A}$

Shunt field Cu loss, $W_{sh} = 5^2 \times 50 = 1250\ \text{W}$

Armature current, $I_a = 200 + 5 = 205\ \text{A}$

Armature Cu loss, $W_a = 205^2 \times 0.08 = 3362\ \text{W}$

Total Cu loss $= 1250 + 3362 = 4612\ \text{W}$

Total losses $= 1250 + 3362 + 900 = 5512\ \text{W}$

Output power, $P_o = 250 \times 200 = 50,000$ W

Input power, $P_i = \text{Output+losses} = 50,000 + 5512 = 55512$ W

Efficiency, $\eta_i = \dfrac{\text{Output}}{\text{Input}} \times 100 = \dfrac{50000}{55512} = 90\ \%$

Constant losses, $W_c = 1250 + 900 = 2150$ W

At maximum efficiency, $W_c = I_a{}^2 R_a$

$$2150 = I_a{}^2 \times 0.08$$

$$I_a = 163.9 \text{ A}$$

Corresponding load current at maximum efficiency, $I_L = I_a - Ish = 163.9 - 5 = 158.9$ A

Armature Cu loss at max efficiency $= 2150$ W

Maximum efficiency $= \dfrac{50000}{50000 + 2150 + 2150} \times 100 = 92.1\ \%$

Applications of D.C Generators

Separately excited generators

i) These are used for speed control of D.C motors over a large range.

ii) These are used in areas where a wide range of terminal voltage is required

Self excited generators

i) shunt generators :-

i) These are used as exciters for exciting the field of synchronous machines and separately excited D.C generators

ii) These are used for battery charging because it's terminal voltage are almost constant or can be kept constant.

iii) Commonly used in ordinary lighting purposes and power supply purposes.

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ii) series generators:-

i) These are used for series arc lighting

ii) Series incandescent lighting

iii) As a series booster for increasing the voltage across the feeder to compensate the resistance drop of the line. because of their rising characteristic.

iv) Special purposes such as supplying the field current for regenerative breaking of D.C locomotives (railway service).

v) Constant current for welding.

iii) compound generators:-

i) Compound generators are used where constant terminal voltages have to be maintained for different loading conditions.

ii) Cumulatively compound generators:-These are for domestic lighting purposes and to transmit energy over long distance and for heavy power service such as electric railways.

iii) Differential compound generator:- The use of this type of generators is very rare and it is used for special application like arc welding.

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